P07 Collectables

CRE132 worksheet

What you'll make

Coins along the path and gems in the harder spots. Touch one and it vanishes with a sound and a burst of sparkles, and the score in the top right goes up: 1 for a coin, 5 for a gem.

You'll write one script that works for both pickups, and make a particle effect that plays once and tidies itself away. By the end you'll know how a small script can hand the work to the managers (score, sound), and why prefabs make changing every coin in a level a one-click job.

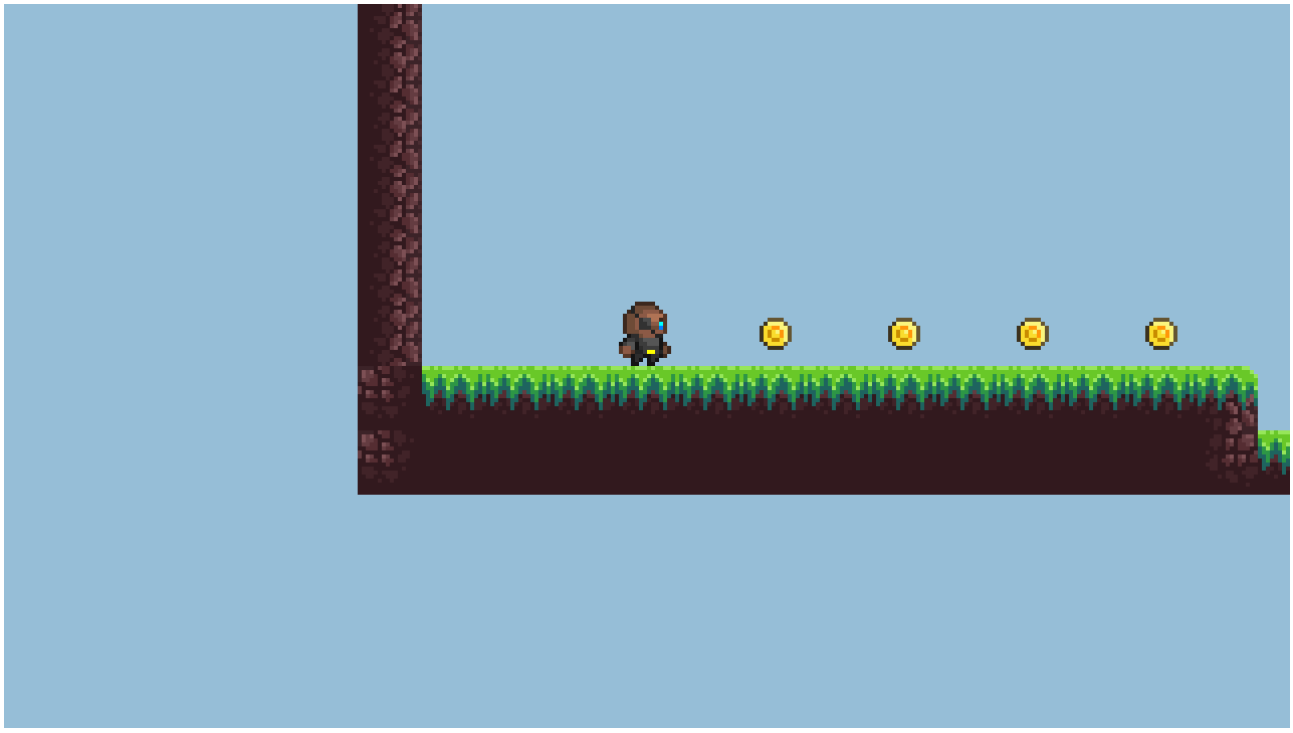


Figure 1: The finished scene: coins along the path, a gem on a high platform and the score in the top right.

Before you start

1. Unzip `CRE132_P07_Collectables_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_P07_Collectables_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Platformer/P07_Collectables`.
3. Press **Play**. You can move (A/D or arrows) and jump (Space), music plays, and the top right shows **Score: 0**. The coins and gems are animated, but you walk straight through them. Press Play again to stop.

The scene has the **Player** (tagged **Player**), a **Collectables** object holding 15 **Coins** and 3 **Gems**, the **GameManager** and **AudioManager** from S02 and S05, and the **HUD Canvas** with the score from S03. Every coin is a copy of the **Coin** prefab, and every gem of the **Gem** prefab.

Making a script: in the Project window, go to the folder, right-click and choose **Create > Scripting > MonoBehaviour Script**. Type the name straight away and press Enter. The name must match the class name in the code exactly, capital letters included.

1. The Collectable script

Make a folder `Assets/_game/Scripts/P07_Collectables` and create a script in it called `Collectable`:

```
// A pickup worth points (a coin, a gem): adds to the score, plays a sound and leaves a
// burst of particles.
using UnityEngine;

public class Collectable : MonoBehaviour
{
    [SerializeField] private int value = 1;
    [SerializeField] private AudioClip pickupSound;

    // A prefab with a Particle System that plays once and removes itself
    // (Stop Action: Destroy).
    [SerializeField] private GameObject pickupEffect;

    private void OnTriggerEnter2D(Collider2D other)
    {
        if (!other.CompareTag("Player"))
        {
            return;
        }

        GameManager.Instance.AddScore(value);

        if (pickupSound != null)
        {
            AudioManager.Instance.PlaySound(pickupSound);
        }
        if (pickupEffect != null)
        {
            Instantiate(pickupEffect, transform.position, Quaternion.identity);
        }

        Destroy(gameObject);
    }
}
```

- **`CompareTag("Player")`** makes sure only the player collects things. Anything else that touches the coin is ignored.
- **The managers do the work.** `GameManager.Instance.AddScore` keeps the score and `AudioManager.Instance.PlaySound` plays the sound. The coin needs no slots for them: `Instance` finds them from anywhere.
- **The `!= null` checks** make the sound and the effect optional, so the script still works with empty slots.
- **`Instantiate`** makes a copy of the effect prefab where the coin was. A particle effect is just another prefab.

2. Make the coins collectable

You'll change the **Coin prefab**, not the coins in the scene. Every coin in the level is a copy of it, so they all get the change.

Add it: in the Project window, open `Assets/_game/Prefabs/Collectables` and double-click **Coin** to open it in Prefab Mode. Its **Circle Collider 2D** already has **Is Trigger** ticked. Click **Add Component > Collectable** and set:

- **Value:** 1
- **Pickup Sound:** click the small circle and pick **GS2_Item_Acquire_1** (from `Assets/_game/Audio/SFX`).
- **Pickup Effect:** leave it empty for now.

Click the arrow at the top left of the Hierarchy to leave Prefab Mode.

Test it: press Play and run through the coins. Each one disappears with a sound, and the score goes up by 1.

3. A burst of sparkles

Now the effect. In the Hierarchy, choose **GameObject > Effects > Particle System** and rename it **Sparkle Gold**. In its **Transform**, set **Rotation X** to **0** (it starts at -90, which points the particles away from a 2D camera).

In the **Particle System** component, set these. For a value with two numbers, click the small arrow to its right and choose **Random Between Two Constants**.

- **Main module** (the one named **Sparkle Gold**): **Duration** 0.5, **Looping** unticked, **Start Lifetime** 0.3 and 0.6, **Start Speed** 2 and 4.5, **Start Size** 0.1 and 0.2, **Start Color** hex `FFD94D` (gold), **Gravity Modifier** 0.6, **Simulation Space** World, **Stop Action** Destroy.
- **Emission: Rate over Time** 0. Under **Bursts**, click **+** and set **Count** to 14.
- **Shape: Shape** Circle, **Radius** 0.1.
- **Color over Lifetime:** tick it, then click the gradient bar. Click above the bar to add an alpha key at **60%** with **Alpha** 255, and set the key at the right end to **Alpha** 0.
- **Renderer: Material Pixel Particle** (from `Assets/_game/Materials/Particles`), **Sorting Layer** Objects, **Order in Layer** 5.

What these do:

- **A Burst of 14 with Rate over Time 0** gives one puff of particles at the start and no more.
- **Not looping, with Stop Action: Destroy:** the effect plays once and then deletes its own GameObject, so used sparkles don't pile up in the Hierarchy.
- **Simulation Space: World** leaves the particles where they were made, even if something moves the effect.
- **Pixel Particle** draws each particle as a plain square, so it matches the pixel art. Colour over Lifetime fades them out at the end instead of popping off.

Make it a prefab: make a folder `Assets/_game/Prefabs/Effects` and drag **Sparkle Gold** from the Hierarchy into it. Then delete **Sparkle Gold** from the Hierarchy: the coin makes its own copies.

Add it: open the **Coin** prefab again and drag **Sparkle Gold** from the Project window into its **Pickup Effect** slot.

Test it: play and collect a coin. A gold puff appears and fades, and nothing called **Sparkle Gold (Clone)** stays in the Hierarchy afterwards.

4. The gem

In `Assets/_game/Prefabs/Effects`, select **Sparkle Gold** and press **Ctrl+D** to duplicate it. Rename the copy **Sparkle Blue**, open it and change **Start Color** to hex `73D9FF`.

Add it: open the **Gem** prefab and add **Collectable** with:

- **Value:** 5
- **Pickup Sound: GS2_Item_Acquire_3**
- **Pickup Effect: Sparkle Blue**

Test it: climb to a gem. It gives 5 points, a different sound and a blue puff. Collect everything for a score of 30.

Check your work

- ☐ A coin gives 1 point, a sound and a gold puff, then disappears.
- ☐ A gem gives 5 points, a different sound and a blue puff.
- ☐ The score in the top right goes up as you collect.
- ☐ No **(Clone)** effects are left in the Hierarchy after the sparkles finish.
- ☐ Compare with `CRE132_P07_Collectables_Finished` if something behaves differently.

Key points

- **One script, many pickups.** Coin and Gem use the same `Collectable` with different values, sounds and effects set in the Inspector.
- **Ask the managers.** The pickup tells `GameManager` and `AudioManager` what happened and lets them do the work; it doesn't need references to them.
- **Edit the prefab, not the copies.** Change the Coin prefab once and every coin in the level changes.
- **Effects are prefabs too.** `Instantiate` one, and **Stop Action: Destroy** cleans it up.

If something goes wrong

- **Walking through a coin does nothing:** you added Collectable to one coin in the scene instead of the prefab, or the Player isn't tagged **Player**. Also check the coin's collider has **Is Trigger** ticked.
- **Only one coin works:** you changed a single coin. Open the prefab (double-click it in the Project window) and add the script there.
- **NullReferenceException on `GameManager.Instance` or `AudioManager.Instance`:** the **GameManager** or **AudioManager** object is missing from the scene.
- **No sparkles appear:** the **Pickup Effect** slot is empty, the Transform's **Rotation X** is still −90, or the Renderer's **Sorting Layer** is behind the level.
- **Sparkle clones stay in the Hierarchy: Stop Action** isn't **Destroy**, or **Looping** is still ticked.
- **The sparkles are soft blurry circles:** the Renderer's **Material** isn't **Pixel Particle**.

Try it

1. Make the gem's sparkle bigger and longer: raise its **Start Size**, **Start Lifetime** and burst **Count**.
2. Change the Coin prefab's **Value** to 2 and play. How many coins in the level changed?
3. Add a heart pickup that heals: start from S01's `HealthPickup` and give it a sound and an effect like the Collectable.
4. Instead of making a new sparkle each time, reuse them with S07's `ObjectPool`.